Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Agent runtime engineer who **built AgentProbe end-to-end** - open-source LangGraph platform delivering exactly what Glean's Agentic Runtime team builds: **multi-turn orchestration** (3-agent system: Orchestrator, RAG Tracer, Auto Fix), **tool calling** with structured execution across agent runs, **model routing** across Anthropic, OpenAI, Groq, Ollama by task and latency, and **session state management** across multi-step agent loops. Built observability into every layer: 15-endpoint FastAPI backend, React dashboard, CI/CD eval gates - the same production reliability and tracing mindset Glean's runtime requires. 8+ years production Python; AWS (S3, EC2, RDS, Glue); Docker microservices at scale. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

EXPERIENCE

George Mason University
Aug 2024 - Aug 2025

Research Associate

Fairfax, Virginia

- Built **regression models (R^2 = 0.977)** in Python and R across WHO and UK Road Safety datasets spanning **190+ countries** - multi-source data collection, transformation, validation, and stakeholder reporting.
- Applied **unsupervised ML (Isolation Forest)** to automate data integrity checks across 3 research pipelines - surfaced anomalies that manual review missed, improving downstream model reliability.
- Designed **Tableau, ggplot2, and Quarto dashboards** for faculty leadership - improved research comprehension across two cohorts by **40%**.

Entrata India (Db Xento Systems)
Oct 2018 - Aug 2024

Software Engineer

Pune, India

- Designed and shipped **20+ production microservices and full-stack modules** (Python, PostgreSQL, JavaScript, RabbitMQ) across **120+ Agile sprints** - **100K+ monthly active users, 99.5% uptime** sustained over 6+ months.
- Containerized **30%+ of the microservices fleet** with Docker, cutting execution time by **50%**; tuned **200+ SQL queries** for a **30% production performance gain** across all client environments.
- Applied **Python, PyTorch, and OpenAI API** to automate data transformation pipelines across **1.6M financial records** - production AI on regulated data with output validation and quality gates.
- Migrated all client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents**; built **5+ internal platform tools** cutting resolution time by **40%**.
- Eliminated **80% of manual operational workload** via Python/PL-SQL automation; enforced TDD/BDD, secure coding standards, and peer review practices across all major releases.

Entrata India (Db Xento Systems)
Jul 2017 - Sep 2018

Associate Software Engineer

Pune, India

- Improved system stability by **35%** by building and optimizing **20+ full-stack property management modules** (Python, PostgreSQL) - **99.5% uptime** and **25% faster report generation**.
- Reduced critical issue backlog by resolving **100+ high-priority bugs**, significantly improving platform reliability for end users.

Entrata India (Db Xento Systems)
Jan 2017 - Jun 2017

Trainee Software Engineer

Pune, India

- Delivered production-ready backend APIs and UI modules from day one using Git-based workflows, secure coding standards, and TDD - immediate contributions across Agile delivery cycles.

PROJECTS

AgentProbe - Agentic AI Testing Platform
In Progress

2024-Present | Tech: Python, LangChain, LangGraph, FastAPI, ChromaDB, Anthropic, OpenAI, Groq, Ollama, React, Docker, GitHub Actions | GitHub: github.com/venkatalakshmiparimala/AgentProbe

- Built a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider model routing (Anthropic, OpenAI, Groq, Ollama) - deployed as a **15-endpoint FastAPI backend** running 100% locally at zero cost.
- Automated agent quality gates by building a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD integration - failing runs blocked automatically before production.
- Delivered end-to-end observability via **React dashboard + ChromaDB RAG pipeline** tracking per-run metrics, regression drift, and retrieval quality across concurrent agent test suites.

AI on FHIR - NLP-Driven Medical Records Query System

2024 | Tech: Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings | GitHub: github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced manual clinical retrieval time by **~60%** by building a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings - HIPAA-compliant RBAC across structured medical records.

CrashClarity - Large-Scale Road Accident Analysis (104K Records)

2025 | Tech: Python, R, scikit-learn, Random Forest, Logistic Regression, ggplot2, Feature Engineering, GitHub Pages | Live: Portfolio

- Achieved **85%+ accuracy** on 104K+ UK accident records by building a Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables - deployed interactive dashboard on GitHub Pages for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, PHP, SQL, R, Bash, HTML/CSS

Agent & AI: LangChain, LangGraph, Multi-Agent Orchestration, Multi-Turn State Management, Tool Calling, Model Routing (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, LLM-as-Judge Evaluation, CI/CD Eval Gates, Prompt Engineering

Backend & Infra: FastAPI, Flask, React, PostgreSQL, MySQL, Oracle, Docker, RabbitMQ, Microservices, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions)

ML & Data: PyTorch, TensorFlow, XGBoost, scikit-learn, Isolation Forest, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Visualization: Tableau, Power BI, ggplot2, Plotly, Quarto **Process:** TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum